

Multiple Linear Regression

TTK 4260: Multivariate Data Analysis and ML

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Learning Objectives

- **Explain** what a coefficient in multiple regression means (partial effect, “holding others fixed”)
- **Identify** when coefficient interpretations are reliable (role of correlation/independence)
- **Correctly include** categorical predictors (dummy encoding + baseline) and interpret them
- **Explain** the effect of scaling on coefficients and on regularization
- **Compare** L0 / L1 / L2 regularization in terms of sparsity, stability, and interpretability

Historical Background

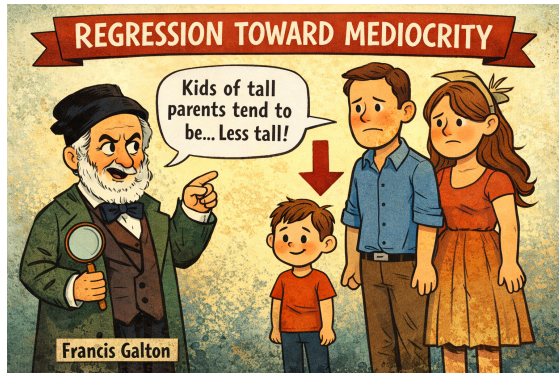
Where Did Regression Come From?

● Early 1800s: Astronomy Problems

- Astronomers needed to reconcile noisy measurements
- Least squares developed to average out errors
- Famous priority dispute: Gauss vs. Legendre

● Late 1800s: The Word “Regression”

- Francis Galton studied parent and child heights
- Children of very tall parents tended to be *less tall*
- He called this “*regression toward mediocrity*”



Statistics began as a way to handle disappointment in data.

Why Multiple Linear Regression?

- Real-world outcomes rarely depend on a single factor
- **House prices are a classic example**
 - Size alone is not enough
 - Two houses with the same size can have very different prices
- Prices are influenced by *multiple variables at the same time*
 - Size, location, age, number of rooms, distance to city center, . . .
- Statisticians eventually accepted an uncomfortable truth:
 - **Reality is multivariate**

From Single to Multiple Linear Regression

Single Predictor Model

Using only house size:

$$y = \theta_0 + \theta_1 x_1$$

Limitation:

- Ignores other important factors
- Leads to biased or noisy predictions

Multiple Linear Regression

Include all relevant features:

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_p x_p + \varepsilon$$

House price features:

- Size
- Number of bedrooms
- Age of the building
- Location / neighborhood
- Distance to city center

Same line-fitting idea — just admitting houses are complicated.

Multiple Regression Model

The Multiple Linear Regression Model

Model Statement

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_p x_p + \varepsilon$$

In matrix notation:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$$

where:

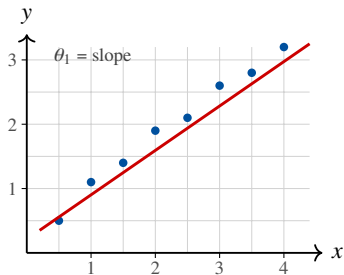
- $\mathbf{y}$: $m \times 1$ vector of responses
- $\mathbf{X}$: $m \times (p + 1)$ design matrix (includes intercept column)
- $\boldsymbol{\theta}$: $(p + 1) \times 1$ vector of coefficients $[\theta_0, \theta_1, \dots, \theta_p]^T$
- $\boldsymbol{\varepsilon}$: $m \times 1$ vector of errors

Same least squares principle:

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|^2 = \min_{\boldsymbol{\theta}} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

What Changes from Simple → Multiple Regression?

Simple Linear Regression Model: $y = \theta_0 + \theta_1 x + \varepsilon$

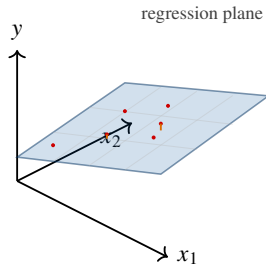


Interpretation:

- θ_1 : change in y per unit change in x
- Clear and unambiguous

Multiple Linear Regression Model:

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \varepsilon$$



Interpretation:

- θ_1 : change in y per unit x_1 , **holding x_2 fixed**
- θ_2 : change in y per unit x_2 , **holding x_1 fixed**
- Requires care!

Understanding “Holding Others Fixed”

Partial Regression Coefficient

θ_j represents the change in y for a one-unit increase in x_j , **while all other predictors remain constant.**

Example: House Price Model

$$y = \theta_0 + \theta_1 \cdot \text{Size} + \theta_2 \cdot \text{Age} + \theta_3 \cdot \text{Bedrooms} + \varepsilon$$

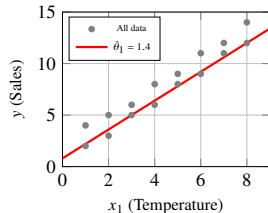
Interpretation:

- $\theta_1 = 3000 \text{ kr/m}^2$: Each additional square meter increases price by 3000 kr, *for houses of the same age and number of bedrooms*
- $\theta_2 = -50000 \text{ kr/year}$: Each additional year of age decreases price by 50,000 kr, *for houses of the same size and bedroom count*
- $\theta_3 = 200000 \text{ kr/bedroom}$: Each additional bedroom increases price by 200,000 kr, *for houses of the same size and age*

Example: Effect of Adding a Confounder

Model 1: Ignoring x_2

$$y = \theta_0 + \theta_1 x_1 + \varepsilon$$

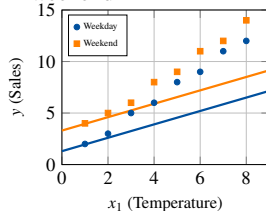


Slope = 1.4 kr/degree

Model 2: Including x_2

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \varepsilon$$

where $x_2 = \begin{cases} 0 & \text{weekday} \\ 1 & \text{weekend} \end{cases}$



Slope = 0.65 kr/degree, Weekend boost = 2.0 kr

Key Insight

Weekends have higher temperatures *and* more customers. Model 1 wrongly attributes everything to temperature. Model 2 separates the effects: true temperature effect is smaller!

Interpretability & Correlated Predictors

Best Case: Orthogonal Predictors

When predictors are uncorrelated:

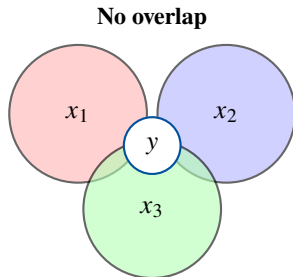
Properties:

- Clean credit assignment
- θ_j captures unique effect
- Stable coefficients
- Easy to interpret

Mathematical property:

$\mathbf{X}^T \mathbf{X}$ is diagonal

Each predictor explains a separate part of y .



Example: Designed Experiments

In controlled experiments, we can design predictors to be orthogonal for clearest interpretation.

Worst Case: Highly Correlated Predictors

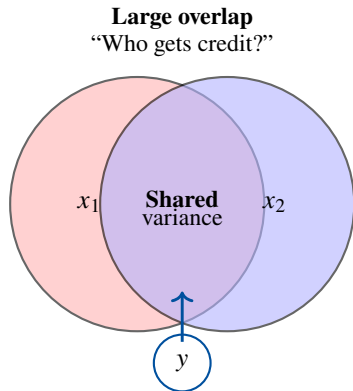
When predictors are correlated:

Problems:

- Shared variance
- Unstable coefficients
- Sign flips possible
- Hard to interpret
- $\mathbf{X}^T \mathbf{X}$ nearly singular

Extreme case:

If $x_2 = 2x_1 \Rightarrow \infty$ solutions

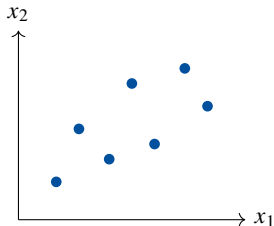


Consequence: Multicollinearity

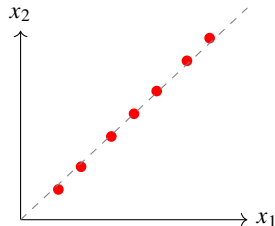
$\mathbf{X}^T \mathbf{X}$ nearly singular $\Rightarrow (\mathbf{X}^T \mathbf{X})^{-1}$ has large entries $\Rightarrow$ high coefficient variance

Visualizing Correlation Effects

Same data, different correlations:



Low Correlation
 $r = 0.3$



High Correlation
 $r = 0.95$

Low correlation:

- Stable θ_1, θ_2
- Clear interpretation

High correlation:

- Unstable θ_1, θ_2
- Difficult interpretation

Board Demo: Who Gets the Credit?

Scenario: Two highly correlated predictors x_1 and x_2

Consider three possible solutions:

Solution A: $\hat{y} = 5 + 10x_1 + 0x_2$

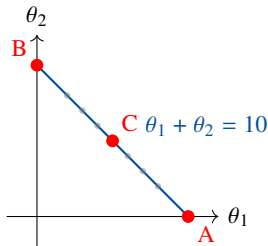
Solution B: $\hat{y} = 5 + 0x_1 + 10x_2$

Solution C: $\hat{y} = 5 + 5x_1 + 5x_2$

If $x_1 \approx x_2$, all three give *similar predictions*!

But the **interpretations** are completely different:

- A: Only x_1 matters
- B: Only x_2 matters
- C: Both matter equally



All points on the line give similar predictions!

Prediction vs. Interpretation

Key Distinction

Prediction accuracy $\neq$ Interpretability

Good for Prediction:

- Correlated predictors? No problem!
- As long as correlations are stable in test data
- Model predicts accurately

Goal: Minimize

$$\sum_{i=1}^m (y_i - \hat{y}_i)^2$$

Many different θ can achieve similar prediction

accuracy.

Bad for Interpretation:

- Can't trust individual θ_j
- Coefficients unstable
- Sign flips possible

Goal: Understand

“What is the effect of x_j ?”

With correlation, this question becomes ambiguous.

Categorical Variables in MLR

Why Categories Aren't Numeric

Problem: How do we include categorical variables like *color*, *city*, or *day of week*?

Common Mistake

Assigning arbitrary numbers:

● Red = 1, Green = 2, Blue = 3

This implies: Blue is “3 times Red” and Green is “halfway between” Red and Blue. **Nonsense!**

Correct approach: Use **dummy variables** (also called *indicator variables* or *one-hot encoding*)

Color	IsRed	IsGreen
Red	1	0
Green	0	1
Blue	0	0

Note: We only need 2 dummies for 3 categories! (More on this next...)

Dummy Variable Encoding

For a categorical variable with k levels:

- Create $k - 1$ dummy variables
- Each dummy is 0 or 1
- One category is the **reference** (baseline)

Example: Day of Week

Day	Mon	Tue	Wed	Thu	Fri	Sat
Monday	1	0	0	0	0	0
Tuesday	0	1	0	0	0	0
Wednesday	0	0	1	0	0	0
Thursday	0	0	0	1	0	0
Friday	0	0	0	0	1	0
Saturday	0	0	0	0	0	1
Sunday	0	0	0	0	0	0

Reference category: Sunday (all dummies = 0)

Interpreting Dummy Variable Coefficients

Model with day-of-week dummies:

$$\begin{aligned}\text{Sales} = & \theta_0 + \theta_1 \cdot \text{Mon} + \theta_2 \cdot \text{Tue} + \theta_3 \cdot \text{Wed} \\ & + \theta_4 \cdot \text{Thu} + \theta_5 \cdot \text{Fri} + \theta_6 \cdot \text{Sat} + \varepsilon\end{aligned}$$

Interpretation:

- θ_0 : Expected sales on **Sunday** (baseline)
- θ_1 : Difference in sales between Monday and Sunday
- θ_2 : Difference in sales between Tuesday and Sunday
- θ_j : Difference between day j and Sunday

Example Values

If $\theta_0 = 1000$, $\theta_1 = 200$, $\theta_2 = 150$:

- Expected Sunday sales: 1000 kr
- Expected Monday sales: $1000 + 200 = 1200$ kr
- Expected Tuesday sales: $1000 + 150 = 1150$ kr

The Dummy Variable Trap

Perfect Multicollinearity

If you include *all* k dummies plus an intercept, you create perfect multicollinearity!

Why? The sum of all dummies always equals 1:

$$\text{Mon} + \text{Tue} + \dots + \text{Sun} = 1 \quad (\text{always})$$

This is the same as the intercept column in $\mathbf{X}$!

Rule: k categories $\Rightarrow k - 1$ dummies

Always omit one category as the reference

$\mathbf{X}^T \mathbf{X}$ becomes singular $\Rightarrow$ no unique solution!

Interactions with Categorical Variables

Different slopes for different groups

Model with interaction:

$$y = \theta_0 + \theta_1 x + \theta_2 D_B + \theta_3 (x \cdot D_B) + \varepsilon$$

where $D_B = \begin{cases} 0 & \text{if Group A} \\ 1 & \text{if Group B} \end{cases}$

Fitted equations by group:

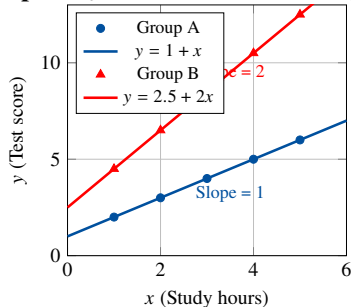
● **Group A** ($D_B = 0$):

$$y = \theta_0 + \theta_1 x$$

● **Group B** ($D_B = 1$):

$$y = (\theta_0 + \theta_2) + (\theta_1 + \theta_3)x$$

Example: $\theta_0 = 1, \theta_1 = 1, \theta_2 = 1.5, \theta_3 = 1$



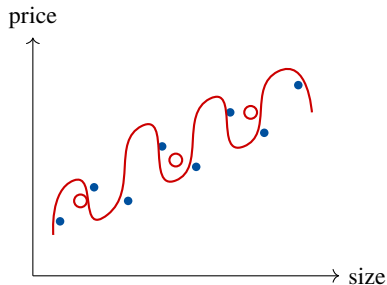
When to Use Interactions

Use interaction terms when the effect of x on y depends on the category. Here, studying helps Group B more than Group A (steeper slope).

Regularization: L2, L1, L0

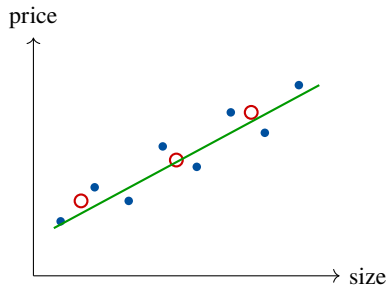
Overfitting

Overly Complex Model



- Fits training data extremely well
- Predictions swing wildly
- Misses new (test) houses

Simpler Model



- Ignores small fluctuations
- Predicts new data well
- Better generalization

A good model fits the signal — not the noise.

When Ordinary Least Squares Starts to Struggle

House price model after adding many features:

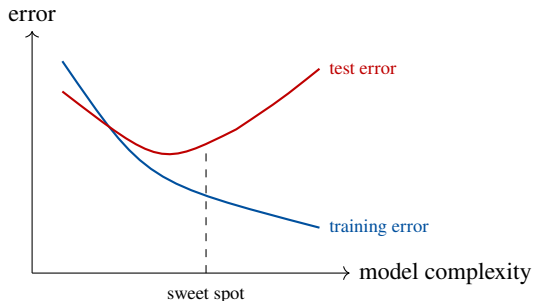
$$\text{Price} = \theta_0 + \theta_1(\text{size}) + \theta_2(\text{bedrooms}) + \theta_3(\text{age}) + \theta_4(\text{distance}) + \theta_5(\text{school quality}) + \dots$$

What students expect:

- More data $\Rightarrow$ better predictions
- More features $\Rightarrow$ smarter model

What actually happens:

- Tiny changes in data $\Rightarrow$ big changes in θ
- Coefficients flip sign unexpectedly
- Model fits training data *too well*



OLS happily fits the noise if we let it.

Why Regularization?

Problems with standard OLS:

- **Overfitting:** Too many predictors $\Rightarrow$ fits noise
- **Multicollinearity:** Correlated predictors $\Rightarrow$ unstable θ
- **Interpretability:** Many small coefficients $\Rightarrow$ hard to understand

Solution: Add a **penalty** to the objective function

$$\min_{\theta} \underbrace{\|\mathbf{y} - \mathbf{X}\theta\|^2}_{\text{Fit to data}} + \underbrace{\lambda \cdot \text{Penalty}(\theta)}_{\text{Complexity control}}$$

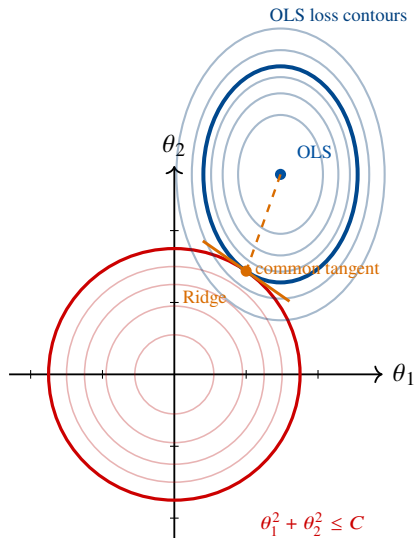
- $\lambda > 0$: Regularization strength (hyperparameter)
- Larger $\lambda \Rightarrow$ simpler model (smaller $|\theta_j|$)
- Smaller $\lambda \Rightarrow$ closer to OLS

L2 Regularization (Ridge): Geometric Intuition

Why circular constraint? L2 penalty: $\theta_1^2 + \theta_2^2 \leq C$

⇒ **circle** in coefficient space

- OLS wants the minimum (center of loss contours)
- Ridge restricts θ to stay inside the circle
- Solution occurs at a **common tangent** point
- Typically **off-axis** ⇒ no exact zeros



L2 Regularization (Ridge)

Objective Function

$$\min_{\theta} \|y - \mathbf{X}\theta\|^2 + \lambda \|\theta\|_2^2 = \min_{\theta} \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \theta_j^2$$

What it does:

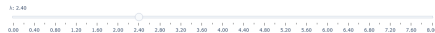
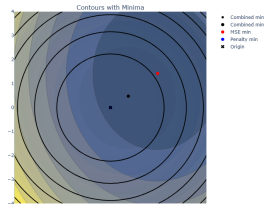
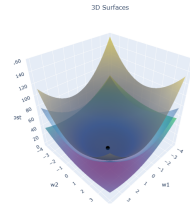
- Shrinks all coefficients toward zero
- Never sets coefficients exactly to zero
- Stabilizes estimates when predictors are correlated

Closed-form solution:

$$\theta_{\text{ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

$\lambda \mathbf{I}$ ensures invertibility and improves conditioning (solves multicollinearity issues).

MSE + L2 Penalty + Combined (with black penalty contours)

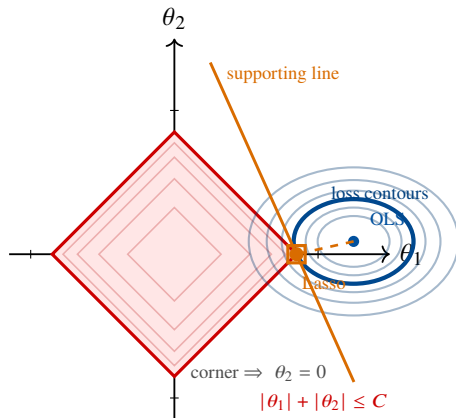


L1 Regularization (Lasso): Geometric Intuition

Why **diamond constraint**? L1 penalty:

$$|\theta_1| + |\theta_2| \leq C$$

- OLS loss contours are smooth ellipses
- L1 constraint has sharp **corners**
- First contact can happen at a corner
- Corner $\Rightarrow$ some $\theta_j = 0$ exactly



Key Property

Lasso solutions often occur at **corners** $\Rightarrow$ exact zeros $\Rightarrow$ automatic feature selection.

L1 Regularization (LASSO)

Objective Function

$$\min_{\theta} \|y - \mathbf{X}\theta\|^2 + \lambda \|\theta\|_1 = \min_{\theta} \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\theta_j|$$

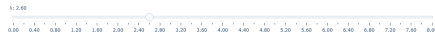
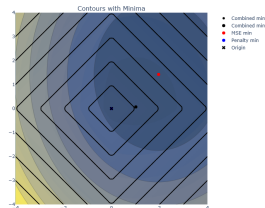
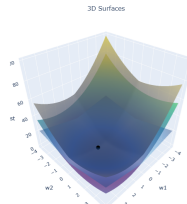
What it does:

- Shrinks coefficients *and* sets many exactly to zero
- Performs automatic **feature selection**
- Produces sparse models (few non-zero θ_j)

Optimization:

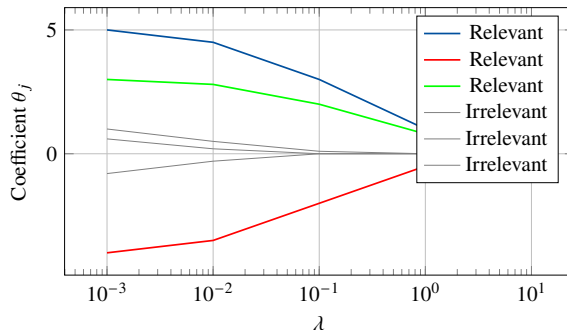
- No closed-form solution
- Convex optimization problem
- Solved iteratively (coordinate descent, LARS, etc.)

MSE + L1 Penalty + Combined (with black penalty contours)



Lasso: Feature Selection Example

Data: 100 samples, 20 predictors, but only 5 are truly relevant.



As λ increases:

- Irrelevant features drop to zero first
- Relevant features persist longer
- Final model is sparse and interpretable

Lasso: Caveat with Correlated Predictors

Problem: When predictors are highly correlated, Lasso may arbitrarily pick one and set others to zero.

Example: two nearly identical predictors

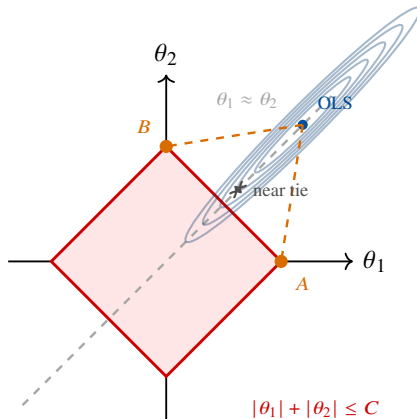
$$x_2 \approx x_1$$

Both are relevant, but Lasso may select:

● $\theta_1 = 5, \theta_2 = 0$ or

● $\theta_1 = 0, \theta_2 = 5$

The choice can be **unstable**: small data changes $\Rightarrow$ different selected variable.

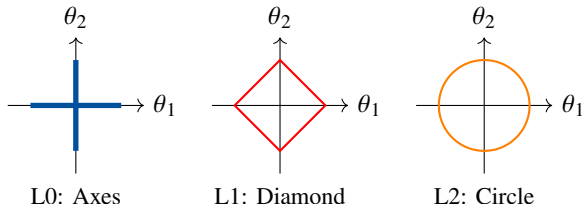


Contours = OLS loss Diamond = L1 constraint A/B = competing corner solutions

Summary of L0, L1, L2

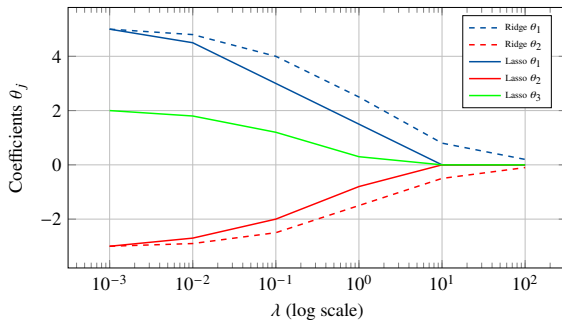
Property	L0	L1 (Lasso)	L2 (Ridge)
Penalty	$\ \theta\ _0 = \sum I(\theta_j \neq 0)$	$\ \theta\ _1 = \sum \theta_j $	$\ \theta\ _2^2 = \sum \theta_j^2$
Sparsity	Exact zeros	Many zeros	No exact zeros
Solution	Combinatorial	Convex (efficient)	Closed-form
Stability	Unstable	Moderate	Very stable
Interpretability	Best	Good	Moderate

Visual comparison:



Regularization Path Example

Effect of increasing λ :



Key observation:

- Ridge (dashed): smooth shrinkage, no exact zeros
- Lasso (solid): hits zero at finite λ

Feature Scaling

Two Perspectives on Scaling

1. Interpretation

- Coefficient units depend on feature units
- Hard to compare magnitudes

Example:

$$\text{Price} = 100 + 3000 \cdot \text{Size}_{\text{m}^2} + 50000 \cdot \text{Size}_{\text{km}^2}$$

Same variable, wildly different θ !

- 3000 kr/m²
- 50,000,000 kr/km²

2. Regularization

- Penalties depend on $|\theta_j|$
- Unfair if scales differ

Example:

$$\text{Penalty} = \lambda(\theta_1^2 + \theta_2^2)$$

If $x_1 \in [0, 1]$ but $x_2 \in [0, 1000]$:

- θ_2 will be ~ 1000 times smaller
- Gets penalized less
- **Unfair!**

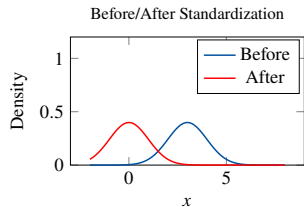
Conclusion

Scaling is **optional** for OLS, but **essential** for regularized regression!

Common Scaling Methods

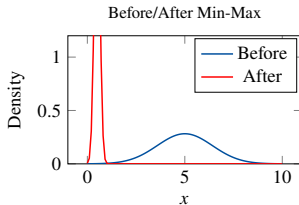
Standardization (Z-score): $x' = \frac{x - \mu}{\sigma}$

- Mean = 0, Std = 1
- Preserves outliers
- Most common



Min-Max Normalization: $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$

- Range = [0, 1]
- Sensitive to outliers
- Useful for bounded data



Critical Rule

Always fit the scaler on training data only, then apply to test data!

Effect on Coefficient Interpretation

Original model (unscaled):

$$\text{Salary} = 30000 + 1000 \cdot \text{Years_Experience} + \varepsilon$$

Interpretation: Each additional year increases salary by 1000 kr. **Standardized model:**

$$\text{Salary} = 30000 + 5000 \cdot \text{Years_Experience}^* + \varepsilon$$

where $\text{Years_Experience}^* = \frac{\text{Years_Experience} - 5}{2}$ (mean=5, std=2) Interpretation: Each *standard deviation* increase (2 years) increases salary by 5000 kr.

Key Point

Standardized coefficients are easier to compare: larger $|\theta_j^*|$ means stronger effect, *regardless of original units*.

Summary

1. Partial Effects & “Holding Fixed”

- θ_j = effect of x_j when other predictors are constant
- Only meaningful if you *can* hold others fixed

2. Correlation Hurts Interpretability

- Shared variance $\rightarrow$ unstable coefficients, sign flips
- Prediction often OK, but interpretation breaks down

3. Categorical Variables

- Use $k - 1$ dummy variables for k categories
- Reference category: all dummies = 0
- Coefficients = difference from reference

Summary (cont.)

4. Scaling Matters for Regularization

- Interpretation: coefficient units change with feature units
- Regularization: penalties unfair without scaling
- **Always scale before L1/L2!**

5. L2 = Stability, L1 = Sparsity, L0 = Best Subset

- Ridge (L2): shrinks all coefficients, very stable
- Lasso (L1): sets many to zero, feature selection
- L0: exact subset selection, but computationally hard

Key Takeaway

Multiple regression is powerful but requires care! Correlation, scaling, and regularization all matter for interpretability and stability.

Thank you!

Questions?

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